

Federated Learning for Internet of Things: Applications, Challenges, and Opportunities

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Abstract—Billions of IoT devices will be deployed in the near future, taking advantage of the faster Internet speed and the possibility of orders of magnitude more endpoints brought by 5G/6G. With the blooming of IoT devices, vast quantities of data that may contain private information of users will be generated. The high communication and storage costs, mixed with privacy concerns, will increasingly be challenging the traditional ecosystem of centralized over-the-cloud learning and processing for IoT platforms. Federated Learning (FL) has emerged as the most promising alternative approach to this problem. In FL, training of data-driven machine learning models is an act of collaboration between multiple clients without requiring the data to be brought to a central point, hence alleviating communication and storage costs and providing a great degree of user-level privacy. We discuss the opportunities and challenges of FL for IoT platforms, as well as how it can enable future IoT applications.

Index Terms—Federated Learning, IoT, Privacy-preserving Machine Learning

I. INTRODUCTION

Recently, Facebook has changed its company name to Meta as it shifts its focus to the metaverse, the universe of virtual worlds that are all interconnected [1]. That brings the collaborative and active learning for the Internet of Things (IoT) back to the spotlight, as a massive number of IoT devices will be deployed to develop an ecosystem that will eventually evolve into the metaverse. However, the fast expansion of IoT networks poses new challenges to the traditional centralized over-the-cloud approaches for data analysis as IoT devices are generating an unprecedented amount of data [2]. On the one hand, centralized approaches no longer fit the 5G/6G era due to the extremely high communication and storage overhead (e.g., high-frequency data from high-volume time-series sensors such as video cameras or Lidar sensors) for pooling data from many IoT devices. Cisco has estimated that nearly 850 ZB of data will be generated by all the people, machines, and things on the network edge by the end of 2021 [3], but annual global data center IP traffic will only reach 20.6 ZB in this year. On the other hand, the data being collected is increasingly viewed as threatening user privacy. With the centralized over-the-cloud approaches, user data will be shared between or even sold to various companies, violating privacy rights and negatively affecting data security, further driving public distrust with data-driven applications. Therefore, an innovative data management approach for data-driven learning and inference-based applications is needed for efficiency and to alleviate privacy concerns.

Federated learning (FL), a nascent field, has been proposed as a distributed privacy-preserving solution to addressing this pressing need [4]. In FL, training of the machine learning

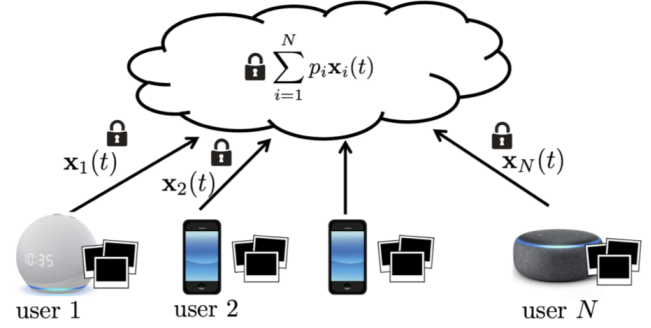


Fig. 1: Federated Learning for IoT Devices.

models needed for data-driven applications is an act of collaboration between multiple clients without centralizing the data of the clients so as to preserve user privacy. The distributed nature of FL is a good match to the IoT network where each edge device in the IoT network is an individual client. Since the raw data of each device will not be exposed to others, FL is an effective way to address the above challenges, particularly in the IoT domain, where the sensors may directly sense or infer data about users that contains sensitive personal information.

In this article, we explain the advantages that FL brings to the IoT domain. We will also discuss some of the most important IoT applications enabled by these advantages. Lastly, we will discuss some of the outstanding challenges when applying FL to IoT applications, and the opportunities in tackling these challenges. We hope this article can act as an enabler to inspire new research at the intersection of federated learning and IoT.

II. WHY FEDERATED LEARNING FOR IoT?

The term federated learning was first introduced in 2016 by McMahan et al. [5]. As shown in Figure 1, FL is a framework for collaborative training of global machine learning models from multiple datasets that are distributed over separate clients, whereas the datasets are not shared among clients during the training phase. There are generally two main roles inside an FL framework: clients holding their own datasets for the training, and the server that arranges the entire training process and generates the global model without accessing any local datasets. During the FL training, the server first initializes the model weights and hyperparameters and distributes the model to individual clients. After receiving the model information sent by the server, each client starts to train the model using its local dataset and then sends its updated local information,

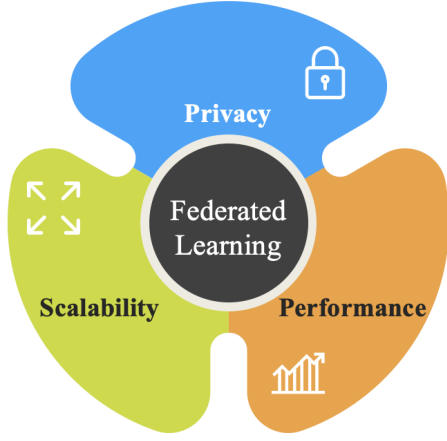


Fig. 2: Advantages of Federated Learning for IoT Devices.

such as the model weights or the gradients, to the server for model aggregation. FedAvg is one of the most popular aggregation functions [6], which generates the global model by equally averaging weights or gradients of each model. After model aggregation, the server distributes the global model to all clients for the next round of training.

The privacy-preserving, collaborative, and distributed nature of FL brings a number of key advantages for IoT applications (Figure 2) as follows:

- **Preserving the Privacy of User Data:** In an ideal FL scenario, each IoT device in the system would learn nothing more than the information needed to play their role. The raw data never leaves the devices during the process and only the updates of the model will be sent to the central server, which minimizes the risk of personal data leakage.
- **Improving Model Performance:** Due to device constraint, single IoT devices may not have sufficient data to learn a high-quality model by itself. Under the FL framework, all the IoT devices can collaboratively train a high-quality model such that each participant could benefit from learning data collected by others beyond its own data. Thus, FL is an effective mechanism to enhance the model performance that each individual device cannot achieve by itself.
- **Flexible Scalability:** The distributed nature of FL is able to leverage the constrained computation resources located at multiple IoT devices across different geographical locations in a parallel manner. By attracting more devices to join the framework, FL enhances the scalability of IoT networks without adding an extra burden on a centralized server due to its distributed learning nature. In addition, within the FL framework, there is no need for the expansive transmission of the raw IoT-collected data. Instead, the communication to the server will mainly be focused on the model updates, which also increases the scalability with regard to communication costs, especially for the low bandwidth IoT networks.

III. KILLER APPLICATIONS

Enabled by the advantages mentioned above, FL has enabled many important IoT applications. In this section, we discuss some of the most important ones (Figure 3).

A. Healthcare

As IoT devices become more pervasive in individuals' daily lives, the privacy of the collected data becomes a significant matter. An example to illustrate privacy concerns is IoT E-health. Nowadays, smart wearable devices are used to monitor the health status of patients such as heartbeat, blood pressure, and glucose level. Compared to the other types of data, personal healthcare data is most sensitive to privacy and highly restricted to the government regulation for any kinds of data sharing [7]. To moderate the above challenges, FADL [8] first proposes a federated training framework that allows each hospital to participate in learning part of the model using its own medical data source. Yuan *et al.* [9] presents a similar co-operative healthcare framework enabled by FL among medical IoT devices. Very recently, the potential of FL has also been discovered during the fight with the COVID-19 pandemic. Liu *et al.* [10] brings an FL-based system that allows hospitals to cooperate in training the models for identifying CT scans of COVID-19 patients without transmitting raw data, which breaks the data isolation problem for IoT E-health.

B. Cybersecurity

With the expansion of the IoT networks, the potential traffic volume of IoT-based DDoS attacks is reaching unprecedented levels, as witnessed during the Mirai botnet attack leveraging infected webcams and home routers [11]. Attacks such as this have raised awareness of the need for IIoT risk assessment and security, e.g. in fields such as healthcare. FL can offer an alternative approach for IoT cybersecurity by protecting the network from malicious attacks as close as possible to the edge devices. DIoT [12] is the first system to employ an FL approach to anomaly-detection-based intrusion detection in gateways to IoT devices. IoTDefender [13] is another similar framework, which obtains personalized models on each local device by fine-tuning the global model trained with FL to filter the malicious traffic. Recently, the FedML Ecosystem [14]–[18] proposes the first FL platform designed specifically for IoT devices and the FedDetect algorithm to catch anomalous traffic. Their experiments on the Raspberry Pi demonstrate the efficiency and feasibility of deploying FL on IoT devices.

C. Smart Home

Smart home systems enabled by consumer IoT devices have achieved great popularity in the last few years as they improve the comfort and quality of life for the residents [19]. The wireless smart IoT home devices, such as smart bulbs, smart doorbells, and smart cameras, are capable of communicating with each other and controlled remotely by smartphones and microcontrollers. People now tend to rely on smart IoT gateways like Amazon Alexa and Apple HomePod to control the house hands-free. While enjoying the convenience brought

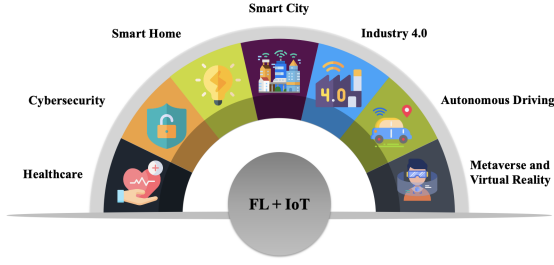


Fig. 3: Applications of Federated Learning for IoT Devices.

by these IoT devices, we could not ignore the essence of privacy-preserving, especially in our own homes. For example, the leakage of the personal voice data for the voice control gives hackers an opportunity to create the imitation. Many works have implemented the FL to address the above challenge. Researchers from Apple have proposed a federated learning model with differential privacy to improve the on-device speaker verification while reducing the risk of raw data leakage [20]. Also, SecWIR [21] is proposed as a framework for securing the smart home IoT communications via Wi-Fi routers.

D. Smart City

IoT-enabled smart cities are bringing significant advancements by making city operations efficient while improving the quality of life for citizens [22]. Various IoT devices enable city managers to control the physical objects in real-time and provide intelligent information to citizens in terms of the traffic system, transportation, public safety, healthcare, smart parking, and smart agriculture, and so on. For example, some researchers propose a dynamic dispatcher for a fleet of electric buses based on the real-time traffic and weather data [23]. Due to the concerns of data privacy, smart infrastructures are moving to compute resources close to where data reside, which makes FL framework suitable for deployment. Distream [24] represents the first distributed framework that enables workload-adaptive live video analytics under the smart camera-edge cluster architecture, which represents a step towards turning the envisioned large-scale live video analytics with privacy-preserving into reality.

E. Industry 4.0

The rapid development in the Industrial Internet of Things (IIoT) brings several advances in information technology applications for the manufacturing field. The concept of Industry 4.0, also known as the fourth industrial revolution, has been proposed based on the emergence of significance for the inter-connectivity based on IIoT and the access to real-time data [25]. With unprecedented connectivity, Industry 4.0 will bring greater insight, control, and data visibility for the supply chain in many industries. However, there are still some real-world problems challenging the deployment of Industry 4.0. First, the amount of data generated from a single factory may not be sufficient enough for training a reliable model comprehensively. Second, the data collected by the IIoT devices is highly related to the commercial value, which makes

privacy-preserving important. For example, eavesdroppers may infer the capacity for manufacture from its electricity usage for industrial IoT users. To address the above challenges, Lu *et al.* [26] proposed a blockchain and FL-based data-sharing framework in IIoT. Its empirical results demonstrate the efficiency of FL usage in IIoT.

F. Autonomous Driving

Along with the advancement of the Vehicular IoT, autonomous driving system is gradually getting close to the ordinary living. For a reliable self-driving system, it needs frequent real-time communication with multi-access communication environment. Also, the spatial and temporal changes of vehicular environment require an intelligent approach that can evolve with the change of environment. For the traditional centralized-over-cloud method, the driving system need to transmit large amount of raw data to the server, which would cause potential privacy leakage. The communication delay triggered the large-size data transmission may also let the driving system could not respond to the real-time spatial changes for the vehicle movement precisely. Using federated learning based on vehicular edge computing for autonomous driving will become a promising direction to mitigate the above challenges. Within FL, each vehicle only needs to transmit a limited size data to the cloud. Also, FL could make the vehicle adopt to real-time local changes more sensitively. In [27], the authors proposed a blockchain-supported federated learning for trustworthy vehicular networks. Within their network, each vehicle will use FL framework to generate a global decentralized model. For the critical cars like police cars or fire cars, the weights will go through the blockchain before to the server to authenticate they are true and authentic. Their simulation results demonstrate that FL could ensure trustworthiness and adherence to delay requirements for vehicles in VNet systems.

G. Metaverse and Virtual Reality

The metaverse is a hypothesized next generation of the internet, providing fully connected, immersive and engaging online 3D virtual experiences through conventional personal computing, as well as virtual and augmented reality devices [28]. In metaverse, users own their avatars and assets and can collaborate and interact with virtual objects and other participants. Metaverse could provide a new platform for business, socialization, entertainment, education, manufacture, and etc. The process towards building the metaverse will require further development of multiple emerging technologies including AI, IoT, AR/VR, edge/cloud compute, blockchain and 5G. To establish the convergence between physical and virtual world, it is essential to create the digital twins of the real objects, environments and assets. By processing data from IoT devices, digital twins could not only replicate the physical appearance and also reflect the real time information such performance and status changes from the physical world. Due to the increasing amount of collected data and data privacy concerns, FL as a distributed collaborative AI approach could further enhance the role of IoT in metaverse scenarios by leveraging the advantages mentioned in section II.

IV. CHALLENGES AND OPPORTUNITIES

To realize the full potential of FL in the applications mentioned above, there still exist several fundamental challenges that need to be addressed. In this section, we describe eight research challenges followed by opportunities that have high promise to address those challenges.

A. Limited Computational Resources

The deployment of FL on IoT networks highly relies on the memory and computational resources at edge devices. However, people often only focus on the ability of IoT devices to collect data, and disregard their limited memory and computational resources, which make it difficult for most IoT devices to finish local computation with large data or complex models [29].

To address this challenge, it is critical to explore lightweight AI algorithms that can be implemented in the resource-constrained FL-IoT environment, as well as optimized resource management methods to facilitate on-device FL training. FedGKT [30] reformulated FL as a group knowledge transfer training algorithm for the large-size model training on resource-constrained edge devices. Facemask [31] is recently proposed as a joint computation and communication-efficient personalized FL framework via heterogeneous masking. In particular, FedMask could improve the communication cost and the computation cost by 34.48x and 2.44x respectively compared to the personalized fedavg framework, which serves as a critical step to mitigate this challenge.

B. Device Heterogeneity

Within cross-device settings, participants under the FL framework have diverse system resources, such as computation and memory resources. As the trend in machine learning is for larger and deeper models, the hardware heterogeneity inside the IoT system poses a challenge for the federated learning framework. As the devices with powerful computation and memory resources, they could train the large models easily while the other devices could only train the smaller models with limited resources. Even for the same model size, the training speed will vary across devices, which may trigger the asynchronous communication problem as we addressed above. An FL framework for IoT should enable graceful adaptation of the data and compute load across different devices based on their resource availability.

C. Limited Networking Bandwidth

Communication overhead is considered one of the major challenges in an FL-based IoT environment. Currently, most of IoT devices communicate using wireless networks whose bandwidth is much less than wired network bandwidth. As the more devices joining into the system, the communication problem is advent when clients possess with different resources allocations. The limited network bandwidth not only makes the communication between clients and server inefficient, but also triggers the presence of straggler clients, which fail to share its local update with the server during the communication round.

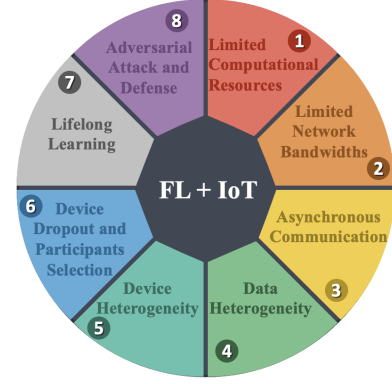


Fig. 4: Challenges of Federated Learning for IoT Devices.

They both serve as the bottlenecks for the performance of FL deployment in the large-scale IoT scenario.

To address this challenge, some key ideas have been proposed in the past few years, such as decentralized training, data compression and participant selections. Mercury [32] recently proposed a sampling-based framework that enables efficient on-device distributed training without compromising the training quality as a new inspiration for solving this challenge.

D. Asynchronous Communication

Currently, most of the FL studies are based on the synchronous update at the server, which means that the server will not start the model aggregation until it receives the information sent from the slowest edge device. However, due to the unbalanced communication abilities and training data distribution, the local training speed varies from device to device, even some participants will temporarily drop off during the training phase, which makes the synchronous update nearly impossible. In the asynchronous FL scenarios, a client could join the active learning group even in the middle of the training progress, which endangers the convergence of the algorithm.

To address this challenge, some researchers proposed asynchronous aggregation scheme with the implementation of coding theory to resist the stragglers in the FL system. In [33], an asynchronous aggregation protocol known as FedBuff has been proposed to mitigate stragglers and enable secure aggregation jointly. Specifically, the individual updates are not incorporated by the server as soon they arrive. Instead, the server will keep receiving local updates in a secure buffer of size K , which is a tunable parameter, and then update the global model when the buffer is full. To extend the limit of buffer size K , [34] is proposed to mitigate the memory limitation for buffered asynchronous aggregation.

E. Data Heterogeneity

In IoT settings, the data collected by different devices can be very different in their characteristics, and various types of devices have different preferences for their data collection, which may create a discrepancy in the local data structure among all the participants in FL. For example, a surveillance

camera will record in real-time (24x7 hours), while the data generated by a doorbell is intermittent. However, the central server could not examine the impact of the data heterogeneity until the training is over.

To address this challenge, we envision that the training quality and speed will be improved if we could determine the heterogeneity and make adjustments accordingly before the training starts. Another popular direction for this challenge is to reduce the model bias via efficient regulations for the training.

F. Device Dropout and Participant Selection

Due to the heterogeneity of data collected and the diversity of IoT devices along with their deployment settings, not every IoT device could effectively contribute to the global model under the FL framework. Especially under large-scale IoT systems, millions of devices will be operated under the same network. Not all of them will be simultaneously available for FL updating. Also, their heterogeneous data distribution may lead to a sub-optimal final result.

To address this challenge, Oort [35] is proposed as a guided participant selection algorithm to address the above challenge. Oort enforces the requirements on the distribution of participant data while improving the duration of federated testing by cherry-picking clients to improve the overall training efficiency.

G. Lifelong/Continual Learning

IoT sensing devices will, by their very nature, continuously collect new data, which will be used to update the model for lifelong or continual learning. While keep providing customized services, the continued model update training poses a new challenge for resource-limited IoT devices. On one hand, as most of the edge devices are memory-limited, their computation resources are not sufficient enough to handle both model inference and training. On the other hand, the energy-constrained nature of IoT devices also makes edge devices could not distribute sufficient resources for the efficient continue learning process. Furthermore, lack of sufficient memory to keep past collected data, may accelerate catastrophic forgetting [36] in continual FL.

To address this challenge, we envision that the lightweight machine learning engine is needed to reduce the memory consumption for on-device training. For example, FedGKT [30], MEST [37], and TinyFL [38] are potential methods to reduce the training memory footprint for efficient on-device learning. To avoid catastrophic forgetting we also envision the use of clustering approaches that identify and store few core datasets from each time interval or leveraging IoT “Hubs” that can store non-sensitive/public datasets to inject memory in the training system. Approaches that can detect temporal distribution shifts at each IoT node to determine when to update the model would also be needed.

H. Adversarial Attack and Defense

In the real-world deployment, the prevalence of the IoT devices also poses an attractive target for adversaries seeking to

launch attacks, such as phishing, identity theft, and distributed denial of service (DDoS). Even though these attacks could be easily defended by installing security patches, many IoT devices lack the requisite computation resources to do so. It is crucial for the IoT system to detect the malicious or broken IoT devices that will ruin the model training with limited resource.

To address this challenge, one of the promising directions is to implement a lightweight security protocol in the IoT system for the detection of broken and malicious devices. For example, SecWIR [21] is proposed as a framework for securing the smart IoT communications via Wi-Fi routers with embedded intelligence for the home-based IoT system.

V. CONCLUDING REMARKS

Federated learning has become an important privacy-preserving for machine learning paradigm. Its distributed nature makes it well suited for deployment in many Internet of Things applications. In this article, we have summarized the advantages and elaborated on some important applications of federated learning for IoT. We have also identified and detailed some key research challenges for the future development of FL on IoT. We hope this article could attract more attention to the federated learning field and inspire creativity and innovation needed to build the FL-IoT ecosystem.

REFERENCES

- [1] I. Meta Platforms, “Welcome to meta,” 2021. [Online]. Available: <https://about.facebook.com/meta/>
- [2] M. Zhang, F. Zhang, N. Lane, Y. Shu, X. Zeng, B. Fang, S. Yan, and H. Xu, “Deep Learning in the Era of Edge Computing: Challenges and Opportunities,” in *Book chapter in Fog Computing: Theory and Practice*, Wiley, 2020.
- [3] Cisco, “Cisco annual internet report (2018–2023) white paper,” 2021. [Online]. Available: <https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>
- [4] P. Kairouz, H. B. McMahan, B. Avent, A. Bellet, M. Bennis, A. N. Bhagoji, K. Bonawitz, Z. B. Charles, G. Cormode, R. Cummings, R. G. L. D’Oliveira, S. Y. E. Rouayheb, D. Evans, J. Gardner, Z. Garrett, A. Gascón, B. Ghazi, P. B. Gibbons, M. Gruteser, Z. Harchaoui, C. He, L. He, Z. Huo, B. Hutchinson, J. Hsu, M. Jaggi, T. Javidi, G. Joshi, M. Khodak, J. Konečný, A. Korolova, F. Koushanfar, O. Koyejo, T. Lepoint, Y. Liu, P. Mittal, M. Mohri, R. Nock, A. Özgür, R. Pagh, M. Raykova, H. Qi, D. Ramage, R. Raskar, D. X. Song, W. Song, S. U. Stich, Z. Sun, A. T. Suresh, F. Tramèr, P. Vepakomma, J. Wang, L. Xiong, Z. Xu, Q. Yang, F. X. Yu, H. Yu, and S. Zhao, “Advances and open problems in federated learning,” *Found. Trends Mach. Learn.*, vol. 14, pp. 1–210, 2021.
- [5] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, “Communication-Efficient Learning of Deep Networks from Decentralized Data,” in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, A. Singh and J. Zhu, Eds., vol. 54. PMLR, 20–22 Apr 2017, pp. 1273–1282. [Online]. Available: <https://proceedings.mlr.press/v54/mcmahan17a.html>
- [6] H. B. McMahan, E. Moore, D. Ramage, and B. A. y Arcas, “Federated learning of deep networks using model averaging,” *CoRR*, vol. abs/1602.05629, 2016. [Online]. Available: <http://arxiv.org/abs/1602.05629>
- [7] B. C. Drole, J. S. Marwaha, B. Hyatt, P. E. Blazar, and S. D. Lifchez, “Electronic communication of protected health information: Privacy, security, and hipaa compliance,” *The Journal of Hand Surgery*, vol. 42, no. 6, pp. 411–416, 2017. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S036350231730477X>
- [8] D. Liu, T. Miller, R. Sayeed, and K. D. Mandl, “Fadl: Federated-autonomous deep learning for distributed electronic health record,” *ArXiv*, vol. abs/1811.11400, 2018.

- [9] B. Yuan, S. Ge, and W. Xing, "A federated learning framework for healthcare iot devices," *ArXiv*, vol. abs/2005.05083, 2020.
- [10] B. Liu, B. Yan, Y. Zhou, Y. Yang, and Y. Zhang, "Experiments of federated learning for covid-19 chest x-ray images," *ArXiv*, vol. abs/2007.05592, 2020.
- [11] E. Bertino and N. Islam, "Botnets and internet of things security," *Computer*, vol. 50, no. 2, pp. 76–79, 2017.
- [12] T. Nguyen, S. Marchal, M. Miettinen, H. Fereidooni, N. Asokan, and A.-R. Sadeghi, "Dfot: A federated self-learning anomaly detection system for iot," 07 2019, pp. 756–767.
- [13] Y. Fan, Y. Li, M. Zhan, H. Cui, and Y. Zhang, "Iotdefender: A federated transfer learning intrusion detection framework for 5g iot," in *2020 IEEE 14th International Conference on Big Data Science and Engineering (BigDataSE)*, 2020, pp. 88–95.
- [14] T. Zhang, C. He, T. Ma, L. Gao, M. Ma, and S. Avestimehr, "Federated learning for internet of things: A federated learning framework for on-device anomaly data detection," *ArXiv*, vol. abs/2106.07976, 2021.
- [15] C. He, A. Shah, Z. Tang, D. Fan, A. N. Sivashunmugam, K. Bhogaraju, M. Shimpi, L. Shen, X. Chu, M. Soltanolkotabi, and S. Avestimehr, "Fedcv: A federated learning framework for diverse computer vision tasks," 2021.
- [16] C. He, K. Balasubramanian, E. Ceyani, C. Yang, H. Xie, L. Sun, L. He, L. Yang, P. S. Yu, Y. Rong *et al.*, "Fedgraphnn: A federated learning system and benchmark for graph neural networks," *arXiv preprint arXiv:2104.07145*, 2021.
- [17] B. Y. Lin, C. He, Z. Zeng, H. Wang, Y. Huang, M. Soltanolkotabi, X. Ren, and S. Avestimehr, "Fednlp: A research platform for federated learning in natural language processing," *ArXiv*, vol. abs/2104.08815, 2021.
- [18] C. He, S. Li, J. So, X. Zeng, M. Zhang, H. Wang, X. Wang, P. Vepakomma, A. Singh, H. Qiu, X. Zhu, J. Wang, L. Shen, P. Zhao, Y. Kang, Y. Liu, R. Raskar, Q. Yang, M. Annavaram, and S. Avestimehr, "FedML: A Research Library and Benchmark for Federated Machine Learning," in *Conference on Neural Information Processing Systems (NeurIPS) Federated Learning Workshop*, 2020.
- [19] Y. Jie, J. Y. Pei, L. Jun, G. Yun, and X. Wei, "Smart home system based on iot technologies," in *2013 International conference on computational and information sciences*. IEEE, 2013, pp. 1789–1791.
- [20] F. Granqvist, M. S. Seigel, R. C. van Dalen, Á. Cahill, S. Shum, and M. Paulik, "Improving on-device speaker verification using federated learning with privacy," in *INTERSPEECH*, 2020.
- [21] X. Lei, G.-H. Tu, C.-Y. Li, T. Xie, and M. Zhang, "SecWIR: Securing Smart Home IoT Communications via Wi-Fi Routers with Embedded Intelligence," in *ACM International Conference on Mobile Systems, Applications, and Services (MobiSys)*, 2020.
- [22] H. Arasteh, V. Hosseini, V. Loia, A. Tommasetti, O. Troisi, M. Shafie-khah, and P. Siano, "Iot-based smart cities: A survey," in *2016 IEEE 16th International Conference on Environment and Electrical Engineering (EEEIC)*. IEEE, 2016, pp. 1–6.
- [23] A. Moradipari, N. Tucker, T. Zhang, G. Cezar, and M. Alizadeh, "Mobility-aware smart charging of electric bus fleets," in *2020 IEEE Power Energy Society General Meeting (PESGM)*, 2020, pp. 1–5.
- [24] X. Zeng, B. Fang, H. Shen, and M. Zhang, "Distream: Scaling Live Video Analytics with Workload-Adaptive Distributed Edge Intelligence," in *ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2020.
- [25] L. D. Xu, E. L. Xu, and L. Li, "Industry 4.0: state of the art and future trends," *International Journal of Production Research*, vol. 56, no. 8, pp. 2941–2962, 2018.
- [26] Y. Lu, X. Huang, Y. Dai, S. Maharjan, and Y. Zhang, "Blockchain and federated learning for privacy-preserved data sharing in industrial iot," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 6, pp. 4177–4186, 2020.
- [27] S. Otoum, I. A. Ridhawi, and H. T. Mouftah, "Blockchain-supported federated learning for trustworthy vehicular networks," *GLOBECOM 2020 - 2020 IEEE Global Communications Conference*, pp. 1–6, 2020.
- [28] L.-H. Lee, T. Braud, P. Zhou, L. Wang, D. Xu, Z. Lin, A. Kumar, C. Bermejo, and P. Hui, "All one needs to know about metaverse: A complete survey on technological singularity, virtual ecosystem, and research agenda," *ArXiv*, vol. abs/2110.05352, 2021.
- [29] J. Ploennigs, J. Cohn, and A. Stanford-Clark, "The future of iot," *IEEE Internet of Things Magazine*, vol. 1, no. 1, pp. 28–33, 2018.
- [30] C. He, M. Annavaram, and A. S. Avestimehr, "Group knowledge transfer: Collaborative training of large cnns on the edge," *ArXiv*, vol. abs/2007.14513, 2020.
- [31] A. Li, J. Sun, X. Zeng, M. Zhang, H. Li, and Y. Chen, "FedMask: Joint Computation and Communication-Efficient Personalized Federated Learning via Heterogeneous Masking," in *ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2021.
- [32] X. Zeng, M. Yan, and M. Zhang, "Mercury: Efficient on-device distributed dnn training via stochastic importance sampling," in *The 19th ACM Conference on Embedded Networked Sensor Systems (SenSys'21)*, 2021.
- [33] J. Nguyen, K. Malik, H. Zhan, A. Yousefpour, M. G. Rabbat, M. Malekesmaeili, and D. Huba, "Federated learning with buffered asynchronous aggregation," *ArXiv*, vol. abs/2106.06639, 2021.
- [34] J. So, R. E. Ali, B. Güler, and A. S. Avestimehr, "Secure aggregation for buffered asynchronous federated learning," *ArXiv*, vol. abs/2110.02177, 2021.
- [35] F. Lai, X. Zhu, H. V. Madhyastha, and M. Chowdhury, "Oort: Efficient federated learning via guided participant selection," in *USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2021, pp. 19–35.
- [36] W. Hu, Z. Lin, B. Liu, C. Tao, Z. Tao, J. Ma, D. Zhao, and R. Yan, "Overcoming catastrophic forgetting for continual learning via model adaptation," in *ICLR*, 2019.
- [37] G. Yuan, X. Ma, W. Niu, Z. Li, Z. Kong, N. Liu, Y. Gong, Z. Zhan, C. He, Q. Jin *et al.*, "Mest: Accurate and fast memory-economic sparse training framework on the edge," *arXiv preprint arXiv:2110.14032*, 2021.
- [38] H. Cai, C. Gan, L. Zhu, and S. Han, "Tinyt1: Reduce activations, not trainable parameters for efficient on-device learning," 2020.

VI. BIOGRAPHY SECTION



Tuo Zhang Tuo Zhang (tuozhang@usc.edu) received his B.S. degree in electrical engineering from University of California, Santa Barbara in 2020. He is currently working toward a Ph.D. in Viterbi School of Engineering, University of Southern California. His research interest is in developing trustworthy machine learning algorithms and systems.

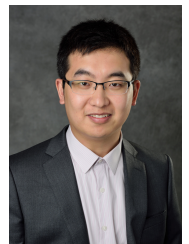


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Chaoyang He Chaoyang He is a Ph.D. Candidate in the CS department at the University of Southern California. His research focuses on distributed/federated machine learning algorithms, systems, and applications. He is advised by Professor Salman Avestimehr (USC) and Mahdi Soltanolkotabi (USC). Previously, He was an R&D Team Manager and Staff Software Engineer at Tencent (2014-2018), a Team Leader and Senior Software Engineer at Baidu (2012-2014), and a Software Engineer at Huawei (2011-2012).

Chaoyang He has received a number of awards in academia and industry, including Best Paper Award at NeurIPS 2020 Federated Learning workshop, Amazon Machine Learning Fellowship (2021-2022), Qualcomm Innovation Fellowship (2021-2022), Tencent Outstanding Staff Award (2015-2016), WeChat Special Award for Innovation (2016), Baidu LBS Group Star Awards (2013), and Huawei Golden Network Award (2012). During his Ph.D. study, he has published papers at ICML, NeurIPS, CVPR, ICLR, MLSys, among others. His homepage: <https://chaoyanghe.com>.



Mi Zhang Mi Zhang (mizhang@msu.edu) is an Associate Professor and the Director of the Machine Learning Systems Lab at Michigan State University. He received his Ph.D. from University of Southern California and B.S. from Peking University. Before joining MSU, he was a postdoctoral scholar at Cornell University. His research lies at the intersection of mobile/edge/IoT systems and machine learning, spanning areas including On-Device AI, Automated Machine Learning (AutoML), Federated Learning, Systems for Machine Learning, Machine Learning

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Dr. Zhang has received a number of awards for his research. He is the 4th Place Winner of the 2019 Google MicroNet Challenge, the Third Place Winner of the 2017 NSF Hearables Challenge, and the champion of the 2016 NIH Pill Image Recognition Challenge. He is the recipient of seven best paper awards and nominations. He is also the recipient of the National Science Foundation CRII Award, Facebook Faculty Research Award, Amazon Machine Learning Research Award, and MSU Innovation of the Year Award.



Bhaskar Krishnamachari Bhaskar Krishnamachari is Professor of Electrical and Computer Engineering at the USC Viterbi School of Engineering. He is the founding director of the USC Viterbi Center for Cyber-Physical Systems and the Internet of Things. He received his M.S. and Ph.D. in Electrical Engineering from Cornell University in 1999 and 2002 respectively, and his B.E. in Electrical Engineering from The Cooper Union for the Advancement of Science and Art in 1998. His research interests pertain to the design and analysis of algorithms, protocols and applications for the internet of things, distributed computing, blockchain technologies, and networked robotics. He is the recipient of an NSF CAREER Award, the ASEE Terman Award, IEEE-HKN Outstanding Young Electrical and Computer Engineer Award, and several conference best paper awards including at ACM Mobicom and IEEE/ACM IPSN.



A. Salman Avestimehr A. Salman Avestimehr is a Dean's Professor, the inaugural director of the USC-Amazon Center on Secure and Trusted Machine Learning (Trusted AI), and the director of the Information Theory and Machine Learning (vITAL) research lab at the Electrical and Computer Engineering Department of University of Southern California. He is also an Amazon Scholar at Alexa AI. He received his Ph.D. in 2008 and M.S. degree in 2005 in Electrical Engineering and Computer Science, both from the University of California, Berkeley. Prior to that, he obtained his B.S. in Electrical Engineering from Sharif University of Technology in 2003. His research interests include information theory, large-scale distributed computing and machine learning, secure and private computing/learning, and federated learning.

Dr. Avestimehr has received a number of awards for his research, including the James L. Massey Research & Teaching Award from IEEE Information Theory Society, an Information Theory Society and Communication Society Joint Paper Award, a Presidential Early Career Award for Scientists and Engineers (PECASE) from the White House (President Obama), a Young Investigator Program (YIP) award from the U. S. Air Force Office of Scientific Research, a National Science Foundation CAREER award, the David J. Sakrison Memorial Prize, and several Best Paper Awards at Conferences. He has been an Associate Editor for IEEE Transactions on Information Theory and a general Co-Chair of the 2020 International Symposium on Information Theory (ISIT). He is a fellow of IEEE. <https://www.avestimehr.com>.